

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 14 Metabolic Diversity of Microorganisms

14.1 Multiple Choice Questions

- 1) The metabolic diversity of photosynthetic bacteria stems from different
- A) bacteriochlorophylls and pigments they contain.
 - B) chlorophylls they can have and organic compounds they can produce.
 - C) light-harvesting complexes, electron donors, and organic compounds they produce.
 - D) unrelated taxa capable of photosynthesis.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

- 2) Whether an organism is classified as a photoheterotroph or a photoautotroph depends on its
- A) energy source.
 - B) carbon source.
 - C) oxygen requirements.
 - D) carbon and energy sources.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

- 3) In photosynthesis, NADH and NADPH are produced from NAD⁺ and NADP⁺ by
- A) oxidation reactions.
 - B) reduction reactions.
 - C) both oxidation and reduction reactions.
 - D) neither oxidation nor reduction reactions.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

- 4) Bacteriochlorophyll and chlorophyll contain _____ as a cofactor.

- A) iron (II)
- B) iron (III)
- C) magnesium
- D) manganese

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

5) Which of the following is NOT a role of carotenoids?

- A) as accessory pigments
- B) photoprotection
- C) to produce singlet oxygen
- D) to quench toxic oxygen species

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.2

6) Proteomic analysis of a microbial community indicated an abundance of phycobiliproteins. Which phototrophic group is likely active and abundant in this community?

- A) cyanobacteria
- B) eukaryotic phototrophs
- C) green bacteria
- D) prochlorophytes

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.2

7) At some of the lowest light concentrations, _____ can still grow well due to their _____, which effectively harvest photons for energy.

- A) green bacteria / antenna pigments
- B) green bacteria / chlorosomes
- C) purple bacteria / antenna pigments
- D) purple bacteria / chlorosomes

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 11.2

8) Phycoerythrin is

- A) the red phycobiliprotein.
- B) the blue phycobiliprotein.
- C) a type of carotenoid.
- D) a green carotenoid.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.2

9) Two separate photosystems involved in electron flow is a hallmark of

- A) anoxygenic phototrophs.
- B) green sulfur bacteria.
- C) oxygenic phototrophs.
- D) purple bacteria.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.4

10) Membrane vesicles known as chromatophores, which function in photosynthesis, are commonly found in

- A) algae.
- B) green sulfur bacteria.
- C) most autotrophic organisms.
- D) purple phototrophic bacteria.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

11) "Special pair" is the name given to the _____ in the photochemical complex of the purple bacteria.

- A) two bacteriochlorophyll *a* molecules
- B) two bacteriochlorophyll *b* molecules
- C) two quinones
- D) two reaction centers

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.3

12) What will happen to a cyanobacterium that has its photosystem II (PSII) blocked?

- A) Additional electron acceptors, such as NADP^+ , will be required to oxidize oxygen and overcome the lost PSII process.
- B) Anoxygenic photosynthesis only using photosystem I (PSI) could occur by using cyclic photophosphorylation and an alternative electron donor such as H_2S .
- C) It will die from being unable to obtain energy for photosynthesis.
- D) Photons will generate excessive reactive oxygen species and the cyanobacterium will die as a consequence.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.4

13) Which group of microorganisms would the Calvin cycle LEAST likely be found in?

- A) purple *Bacteria*
- B) chemolithotrophic *Bacteria*
- C) cyanobacteria
- D) anoxygenic *Chloroflexus*

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.5

14) The process by which electrons from the quinone pool are forced against the thermodynamic gradient to reduce NAD^+ to NADH is called reverse

- A) proton motive force.
- B) reduction.
- C) electron transport.
- D) energy flow.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.3

15) The path of electron flow in oxygenic phototrophs is referred to as the _____ scheme.

- A) E
- B) S
- C) Q
- D) Z

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.4

16) Plastocyanin is a

- A) membrane-bound sac found in certain bacteria.
- B) photosynthetic pigment found in some bacteria.
- C) copper-containing protein in photosystem II that donates electrons to photosystem I.
- D) blue-green bacterium known for its unusual photoreactive complex.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.4

17) The Calvin cycle

- A) is responsible for the fixation of CO_2 into cell material.
- B) utilizes both NAD(P)H and ATP.
- C) requires both ribulose biphosphate carboxylase and phosphoribulokinase.
- D) uses CO_2 , NAD(P)H , and ATP to make biomass with ribulose biphosphate carboxylase and phosphoribulokinase.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.5

- 18) Regarding CO₂ fixation mechanisms in the autotrophic green sulfur bacteria,
A) *Chlorobium* uses the reverse citric acid cycle, and *Chloroflexus* uses the hydroxypropionate pathway.
B) *Chlorobium* uses the hydroxypropionate pathway, and *Chloroflexus* uses the reverse citric acid cycle.
C) both *Chlorobium* and *Chloroflexus* use the reverse citric acid cycle.
D) both *Chlorobium* and *Chloroflexus* use the hydroxypropionate pathway.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.5

- 19) In most cases, the final product of sulfur oxidation is
A) hydrogen sulfide.
B) elemental sulfur.
C) sulfate.
D) thiosulfate.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.9

- 20) Identifying carboxysomes in a bacterium suggests it
A) contains the reverse citric acid cycle.
B) has a deficient Calvin cycle and accumulated CO₂.
C) is in a carboxylic acid rich environment and is storing excess quantities for potentially harsh conditions.
D) will use the Calvin cycle to convert the concentrated into biomass.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.5

- 21) Ferrous iron (Fe²⁺) oxidation generally occurs in environments with
A) alkaline conditions.
B) high H⁺ concentrations.
C) high oxygen content.
D) little or no light present.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.10

22) Alternative autotrophic routes to the Calvin cycle such as the reverse citric acid cycle and the hydroxypropionate pathway are unified in their requirement for

- A) CO₂.
- B) coenzyme A.
- C) NAD(P)H.
- D) organic compound(s) formed.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.5

23) The aerated upper layer of soil is likely to have _____ concentrations of H₂ for aerobic H₂-oxidizing *Bacteria*, so these bacteria likely _____.

- A) high / thrive in such conditions by not competing with chemoorganotrophs
- B) high / generate important reducing equivalents as byproducts for other microorganisms in the soil
- C) low / do not occur in such habitats
- D) low / will need a chemoorganotrophic way to grow as well

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.8

24) What metabolic advantage do cells have in storing the elemental sulfur byproduct from sulfide oxidation?

- A) The cells avoid producing transport energy waste to secrete the sulfur.
- B) The byproduct serves as an electron reserve for subsequent oxidation.
- C) Sulfur decreases the intracellular acidification for non-acid-tolerant sulfide oxidizers.
- D) The byproduct can be used for other biosynthetic pathways that use sulfur, such as Rieske Fe-S proteins.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.9

25) A cell that lacks sulfite reductase but can still oxidize sulfur for energy could be identified by

- A) adenosine phosphosulfate reductase coupled with substrate-level phosphorylation.
- B) electrons being transferred to cytochrome c prior to shuttling them into the electron transport chain.
- C) identifying an alternative quinone, flavoprotein, or cytochrome.
- D) quantifying the release of sulfate byproduct.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.9

26) The only organisms that perform photosynthesis are ones that produce some form of

A) chlorophyll or bacteriochlorophyll.

B) carotenoids.

C) phycoerythrin.

D) phycocyanin.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 13.1

27) Which of the following are NOT found within the photosynthetic gene cluster of *Rhodobacter* (a purple phototrophic bacterium)?

A) genes encoding reaction center and light-harvesting photocomplexes

B) genes encoding proteins involved in phycobiliprotein biosynthesis

C) genes encoding proteins involved in bacteriochlorophyll biosynthesis

D) genes encoding proteins involved in carotenoid biosynthesis

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.2

28) Anammox is an anaerobic process that generates energy from _____ and forms N₂.

A) ammonia

B) ammonium

C) ammonia and nitrate

D) ammonia and nitrite

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.12

29) What would likely occur if an anammox bacterium was unable to use ladderane lipids?

A) Ammonium rather than ammonia would be used due to ammonia toxicity to other cellular processes within the anammoxosome.

B) It would require a different source for carbon assimilation.

C) Rates of anammox would be considerably slower due to a lack of localized metabolism.

D) Toxic products of the anammox reaction could kill the cell.

Answer: D

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.12

30) Which of the following reactions is classified as a heterofermentation?

A) hexose $\rightarrow$ 2 lactate + 2 H⁺

B) HCOOH $\rightarrow$ H₂ + CO₂

C) glucose $\rightarrow$ lactate + ethanol + CO₂ + H⁺

D) fructose $\rightarrow$ 3 acetate + 3 H⁺

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.20

31) Glucose fermentation by *Clostridium* spp. produces ATP only when acetate and butyrate are produced. Why do these organisms produce acetone and butanol after strong initial activity of generating ATP with acetate and butyrate byproducts?

- A) Acetate and butyrate accumulation creates a deadly acidic environment, which acetone and butanol do not.
- B) Acetate and butyrate are no longer needed for biosynthetic pathways.
- C) Acetone and butanol serve as better sources for NAD(P)⁺ reduction.
- D) Acetone and butanol production is favored for stability to store intracellular carbon sources for potential nutrient poor conditions.

Answer: A

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.21

32) The foul-smelling putrescine byproduct suggests activity of

- A) amino acid fermentation by clostridia.
- B) secondary fermentation.
- C) sulfur-oxidizing bacteria.
- D) syntrophic carbohydrate metabolism.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.21

33) A bacterium that catabolizes a compound by linking ion pumps to establish a proton or sodium motive force for energy

- A) can circumvent substrate-level and oxidative phosphorylation.
- B) makes insufficient energy to grow but enough for cellular maintenance to survive.
- C) requires a second bacterium to cooperatively drive the gradient.
- D) still requires another carbon substrate to provide a carbon source.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.22

34) Which metabolic strategy's existence suggests rapid growth is NOT always the best strategy to survive in the environment?

- A) anaerobic fermentation
- B) disproportionation
- C) methylotrophy
- D) syntrophy

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.23

35) Obligate anaerobes can often use _____ electropositive redox couples than facultative anaerobes, and _____ metabolism is most common in this group as well.

- A) lower / assimilative
- B) lower / dissimilative
- C) higher / assimilative
- D) higher / dissimilative

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.7

36) In *Bacteria*, the most common oxidized form of nitrogen is _____ and of sulfur is _____.

- A) nitrate / sulfate
- B) nitrate / sulfite
- C) nitrite / sulfate
- D) nitrite / sulfite

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.13, 14.14

37) Fermentation often produces CO₂, which can be used by _____ as an electron acceptor for energy.

- A) acetogens
- B) methanotrophs
- C) methanogens
- D) acetogens and methanogens

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.16, 14.17

38) How is ATP made by an acetogen during CO₂ fixation?

- A) Electrons from metal cofactors energize the electron transport chain and drive the proton motive force to activate ATP synthase.
- B) Substrate-level phosphorylation of ADP occurs when coenzyme A is removed from acetyl-CoA.
- C) It is made using substrate level phosphorylation and an ion motive force.
- D) Acetyl CoA is produced from acetate and ATP is produced through substrate level phosphorylation.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.16

39) A researcher lacked the necessary equipment to measure methane production so instead monitored CO₂ concentration as an unknown archaeon grew and produced methane. Why might CO₂ NOT decrease while methane would still increase?

- A) An alternative carbon source such as methanol was used.
- B) CO₂ is not a carbon source used by methanogens.
- C) CO₂ was used as an electron donor but not as a carbon substrate.
- D) Methanogenic *Archaea* containing carboxysomes likely made measuring small losses of CO₂ difficult to conclude.

Answer: A

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.17

40) Methanogens that use methyl-CoM for biosynthesis use _____ as a substrate.

- A) acetate
- B) carbon monoxide
- C) methane
- D) methanol

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.17

41) The serine pathway and ribulose monophosphate pathway can both be used by _____ as a way to incorporate carbon into biomass.

- A) acetogens
- B) anoxygenic hydrocarbon fermenters
- C) methanogens
- D) methylotrophs

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.18

42) What products would be expected to form during anoxic degradation of the seven-carbon compound benzoate following reduction and cleavage of the aromatic ring?

- A) 1 three-carbon compound and 1 four-carbon compound
- B) 1 three-carbon compound and 2 two-carbon compounds
- C) 2 three-carbon compounds and CO₂
- D) 3 two-carbon compounds and CO₂

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.25

43) Organisms that aerobically catabolize methane use the intermediate _____ for biosynthesis and produce _____ when oxidizing the substrate for energy.

- A) CH₂O (formaldehyde) / CO
- B) CH₂O (formaldehyde) / CO₂
- C) HCOO⁻(formate) / CO
- D) HCOO⁻(formate) / CO₂

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.18

44) Which of the following is the best potential reason that anoxic methane-oxidizing *Archaea* have not also acquired the ability to reduce sulfate?

- A) Sulfate is toxic to these archaeans.
- B) Having fewer metabolic capabilities decreases the archaeon's genome size and gives it flexibility to interact with other reducing bacteria, such as nitrate bacteria.
- C) The archaeans are too rare to survive without the bacteria.
- D) The methane-oxidizing *Archaea* are not capable of acquiring this metabolic capability.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.18

45) What metabolism would be favored when there is a lack of electron acceptors?

- A) fermentation
- B) anoxygenic photosynthesis
- C) anoxic ammonia oxidation
- D) acetogenesis

Answer: A

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.19

14.2 True/False Questions

1) The conversion of light into chemical energy is called photoautotrophy.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

2) Some purple bacteria harvest light using bacteriochlorophylls.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

3) Reaction centers ONLY indirectly receive photon energy via light-harvesting molecules.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

4) Chlorosomes are present in purple bacteria but absent in green sulfur and green nonsulfur bacteria.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

5) Carotenoids are hydrophobic accessory pigments and vary widely in the color they can absorb.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.2

6) Each chlorophyll and bacteriochlorophyll type is distinguished by its absorption spectrum.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.2

7) Photooxidation reactions can lead to the production of toxic forms of oxygen and the destruction of the photosynthetic apparatus.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.2

8) The Calvin cycle provides autotrophs the ability to convert inorganic carbon into biomass and generate energy during this process.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.5

9) A bacterium that uses CO₂ as an electron source but CANNOT use it as a carbon source is considered a mixotroph.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.8

10) Phototrophic purple bacteria such as *Rhodobacter* species are capable of nitrogen fixation.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.6

11) Despite being called the reverse citric acid cycle, this pathway is currently identified as the most ancient and widespread autotrophic pathway.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.5

12) Chemolithotrophs that obtain electrons from donors such as sulfide use the same electron transport chains to obtain energy as chemoorganotrophs.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.7

13) Photosystem I is responsible for splitting a water molecule in the first step of oxygenic electron flow.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.4

14) RubisCO converts ribulose biphosphate and CO₂ into two molecules of 3-phosphoglyceric acid (PGA).

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.5

15) Organisms grown with CO₂ as its sole carbon source must have energy in the form of ATP as well as reducing power.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.1

16) Iron-oxidizing bacteria grow better in alkaline environments where protons are less likely to abiotically convert Fe²⁺ into Fe³⁺.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.10

17) Some sulfur-oxidizing bacteria can simultaneously reduce nitrate, which enables them to grow anaerobically.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.9

18) Due to a chemical equilibrium, a syntrophic relationship can be disrupted if the product from the first partner's metabolism is removed too quickly.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.23

19) Because H₂ levels in oxic environments are transient, it is likely that aerobic hydrogen bacteria shift between chemoorganotrophy and chemolithotrophy depending on levels of organic compounds and hydrogen in their habitats.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.8

20) Some anaerobic bacteria not only use organic compounds as a carbon source but can also use them for energy as well.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.19

21) Heterofermentation CANNOT be differentiated from homofermentation based on the compound fermented.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.20

22) A monooxygenase can always be distinguished from a dioxygenase by incorporating only one oxygen atom from O₂ into the substrate rather than both.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.24

23) Reductive dechlorination involves chlorinated organic compounds serving as electron donors and releasing the chloride in inorganic forms.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.15

24) Fermentation of organic compounds can produce acetate.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.19

25) The acetyl-CoA pathway is a primary route for carbon source utilization.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.14

26) When elemental sulfur is provided externally as an electron donor, the organism must attach itself to the sulfur particle because of the extreme insolubility of elemental sulfur.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.9

27) One result of the oxidation of reduced sulfur compounds is a rise in the pH of the medium.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.9

28) Bacteria that are capable of oxidizing both iron and sulfur usually have a strong preference for sulfur oxidation because it yields more energy.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.10

29) Beta-oxidation exclusively removes two carbons at a time to catabolize fatty acids regardless of the carbon chain length.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.24

30) Bacteria that degrade aromatic compounds with reductions steps rather than oxygenase activity prior to ring fission are likely to be anaerobes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.25

14.3 Essay Questions

1) Compare and contrast the prokaryotic and eukaryotic light-gathering machinery function and spatial organization. Why do various chlorophylls show different absorption spectra?

Answer: Both prokaryotes and eukaryotes use chlorophylls to gather light energy. Eukaryotes have chloroplast organelles that house the chlorophylls. Prokaryotes instead have chlorophylls located within other internal membrane systems. For example, cyanobacteria have thylakoid membranes, and green bacteria have chlorosomes. Different chlorophylls (e.g., Chl *a* and Chl *b*) have structural variations at the chemical level, which results in varied capabilities of absorbing different light wavelengths. Consequently, the light that these various chlorophylls absorb also make them transmit (or appear) as different colors too.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.1

2) What is the difference between chlorophyll and bacteriochlorophyll, and which organisms contain each?

Answer: Chlorophylls are used in phototrophs where oxygen is released and can be present in bacteria in addition to eukaryotes. Bacteriochlorophylls are found only in anoxic phototrophs such as purple bacteria and are not present in eukaryotes.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 14.1

3) Explain the Calvin cycle process that produces a full molecule of glucose and regenerates the ribulose biphosphate molecule.

Answer: It takes 6 CO₂ and 6 ribulose biphosphate molecules to go through the Calvin cycle to generate 1 glucose molecule. The first main step in the cycle involves RubisCO creating 3-phosphoglyceric acid (PGA) from CO₂ and ribulose-1,5 biphosphate. ATP and NADPH are then used to convert PGA into glyceraldehyde-3-phosphate (G3P). Finally, ribulose biphosphate is regenerated from G3P. Ultimately, 12 NADPH and 18 ATP are required to synthesize one glucose molecule.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.5

4) In what types of organisms are carboxysomes found and what advantage do they provide for a cell?

Answer: Many prokaryotic autotrophs create inclusions within their cells that harbor large quantities of RubisCO. They are especially advantageous for cells to ensure carboxylation of ribulose-1,5 biphosphate via RubisCO occurs (for carbon assimilation) rather than oxygenation, which requires additional energy input.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.5

5) Describe what occurs when elemental sulfur is provided externally as an electron donor and how energy is obtained.

Answer: Elemental sulfur (S₀) can serve as an electron donor for colorless sulfur bacteria.

Aerobic chemolithotrophy of S₀ uses elemental sulfur, oxygen, and water to form sulfate ions.

The reduced sulfates can then enter the electron transport chain to produce a proton motive force, ultimately driving the production of energy in the form of ATP.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.9

6) Explain why it is unlikely an iron-oxidizing bacterium would thrive in a cold stream with a neutral pH. Also propose an experiment that would test whether iron-oxidizing bacteria are present in the stream.

Answer: In an oxic environment, such as a stream, the iron present in the water will be primarily Fe³⁺. Iron-oxidizing bacteria require Fe²⁺ as an electron source, and this form of iron spontaneously oxidizes into Fe³⁺ by oxygen. Due to such low bioavailability of the necessary Fe²⁺, it would be unlikely iron-oxidizing bacteria would exist in such a system. Experimental approaches will vary but must make clear measuring loss of Fe²⁺ supplemented into the stream water is insufficient evidence of microbial transformation. Iron-oxidizing bacteria could be identified in the stream water by classical enrichment approaches and acidifying the stream water to minimize abiotic iron oxidation. Concomitant growth of an isolate with ferric iron precipitation would be strong evidence such bacteria do occur in the stream.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.10

7) Propose why it would be advantageous for a photosynthetic microorganism to have more than one type of chlorophyll or bacteriochlorophyll.

Answer: Each type of chlorophyll or bacteriochlorophyll absorbs different wavelengths of light, so additional types of chlorophyll can take advantage of more photons to produce even more energy.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.1

8) Under what circumstances does oxygenic photophosphorylation use non-cyclic photophosphorylation and when does it use cyclic photophosphorylation? Also describe what occurs during each process.

Answer: Oxygenic photosynthesis generates ATP via ATPase from a proton motive force, created by electron transport. Electrons are used to reduce NADP⁺ and form NADPH (reducing power); thus electrons are not cycled. When sufficient quantities of NADPH are made, electrons are used to cycle through the electron transport chain, and P680 (PSII) is re-reduced. During cyclic photophosphorylation, electrons are transferred in a closed system where there is no consumption or input of electrons. NADPH is not made; instead, additional ATP is synthesized from the added proton motive force generated.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 14.4

9) Why does an organism that is able to respire both aerobically and anaerobically preferentially undergo aerobic respiration?

Answer: Oxygen as the terminal electron acceptor is preferred over other molecules, because aerobic respiration yields more energy. The glucose produced by aerobes can be directly used as a carbon source, whereas anaerobes use additional energy in the Calvin cycle to obtain a carbon source.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.7

10) Explain why most iron-oxidizing bacteria are obligately acidophilic, and discuss some of the environments where these organisms are found.

Answer: Oxidation of iron at a low pH does not create high yields of energy. However, to compensate for the oxidation of iron, a reduction must also occur to balance the reaction. At a low pH, protons can be subsequently reduced. Bacteria with this metabolic capacity are prevalent in acid-polluted and coal-mining environments.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 14.10

11) Explain why the discovery of iron-oxidizing phototrophs has important implications for both understanding the evolution of photosynthesis and explaining the large deposits of ferric iron (Fe^{3+}) found in ancient sediments on Earth.

Answer: The sediments containing Fe^{3+} predate the existence of photosynthetic cyanobacteria that were responsible for oxygenation of the Earth. It is therefore likely these Fe^{3+} deposits were not formed through spontaneous (abiotic) conversion of Fe^{2+} and O_2 . Instead, it is thought anaerobic chemolithotrophs or anoxygenic phototrophs created these deposits through oxidation of Fe^{2+} into Fe^{3+} .

Bloom's Taxonomy: 1-2: Remembering/Understanding

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